MATH 11 — ELEMENTARY STATISTICS

§7.1

Sampling Distributions

Proportions, means, variances and standard deviations of samples

Today's Goal

After completing this section, you should be able to:

- Understand what a sampling distribution is
- Calculate the sampling distribution for sample proportions
- Calculate the sampling distribution for sample means
- Identify biased and unbiased estimators

Learning Target 11

LT.11 Sampling Distributions

I understand what a sampling distribution is and can calculate the mean of a sampling distribution.

From Individuals to Samples

Pick two numbers from 1, 4, and 6, with replacement, and find their mean.

What means can you get?

$1, 2.5, 3.5, 4, 5,$ and 6.

This means that instead of our random variable representing a randomly selected individual's height (for example), it now represents a statistic for a randomly selected sample of individuals.

Sampling Distribution

Definition 7.1.1

A **sampling distribution** is the probability distribution for a sample statistic.

Distribution of Sample Proportions

Definition 7.1.2

The **sampling distribution of the sample proportion** is the distribution of sample proportions with all samples having the same sample size n taken from the same population.

Distribution of Sample Proportions

The following are true for any sampling distribution of sample proportions:

1. The distribution of sample proportions tends to be normally distributed for large enough sample sizes.
2. The mean of all sample proportions for samples of the same size taken from the same population is equal to the population proportion. In other words, sample proportions *target* the population proportion.

Proportion of Odd Numbers

Suppose that our population consists of the numbers 1, 4, and 6.

(a) Calculate the population proportion of odd numbers.

There is one odd number out of the three numbers in the population. So $p = \frac{1}{3}$.

(b) Take all possible samples of size 2 (with replacement) from this population and calculate the sample proportions of odd numbers.

Sample ($n = 2$)	1,1	1,4	1,6	4,1	4,4	4,6	6,1	6,4	6,6
$\hat{p}$	1.0	0.5	0.5	0.5	0	0	0.5	0	0

Mean of the Sample Proportions

Sample ($n = 2$)	1,1	1,4	1,6	4,1	4,4	4,6	6,1	6,4	6,6
$\hat{p}$	1.0	0.5	0.5	0.5	0	0	0.5	0	0

(c) Calculate the mean of all the sample proportions from part (b).

$\hat{p}$	$P(\hat{p})$	$\hat{p} \cdot P(\hat{p})$
0	$\frac{4}{9}$	0
0.5	$\frac{4}{9}$	$\frac{2}{9}$
1	$\frac{1}{9}$	$\frac{1}{9}$
Total		$\mu_{\hat{p}} = \frac{3}{9} = \frac{1}{3}$

It is important to note that we are working with a probability distribution here. The random variable is $\hat{p}$.

Notice that $p = \mu_{\hat{p}}$.

Distribution of Sample Means

Definition 7.1.4

The **sampling distribution of the sample mean** is the distribution of sample means with all samples having the same size n taken from the same population.

Distribution of Sample Means

The following are true for any sampling distribution of sample means:

1. The distribution of sample means tends to be normally distributed for large enough sample sizes.
2. The mean of all sample means for samples of the same size taken from the same population is equal to the population mean. In other words, sample means *target* the population mean.

Sample Means

Suppose that our population consists of the numbers 1, 4, and 6.

(a) Calculate the population mean.

$$\mu = \frac{1+4+6}{3} = \frac{11}{3}$$

(b) Take all possible samples of size 2 (with replacement) from this population and calculate the sample means.

Sample ($n = 2$)	1,1	1,4	1,6	4,1	4,4	4,6	6,1	6,4	6,6
$\bar{x}$	1.0	2.5	3.5	2.5	4.0	5.0	3.5	5.0	6.0

Mean of the Sample Means

Sample ($n = 2$)	1,1	1,4	1,6	4,1	4,4	4,6	6,1	6,4	6,6
$\bar{x}$	1.0	2.5	3.5	2.5	4.0	5.0	3.5	5.0	6.0

(c) Calculate the mean of all the sample means from part (b).

The table shows the probability distribution for $\bar{x}$.

Notice that $\mu = \mu_{\bar{x}}$.

$\bar{x}$	$P(\bar{x})$	$\bar{x} \cdot P(\bar{x})$
1	$\frac{1}{9}$	$\frac{1}{9}$
2.5	$\frac{2}{9}$	$\frac{5}{9}$
3.5	$\frac{2}{9}$	$\frac{7}{9}$
4	$\frac{1}{9}$	$\frac{4}{9}$
5	$\frac{2}{9}$	$\frac{10}{9}$
6	$\frac{1}{9}$	$\frac{6}{9}$
Total		$\mu_{\bar{x}} = \frac{33}{9} = \frac{11}{3}$

Distribution of Sample Variances

Definition 7.1.7

The **sampling distribution of the sample variance** is the distribution of sample variances with all samples having the same size n taken from the same population.

Distribution of Sample Variances

The following are true for any sampling distribution of sample variances:

1. The distribution of sample variances tends to be skewed right.
2. The mean of all sample variances for samples of the same size taken from the same population is equal to the population variance. In other words, sample variances *target* the population variance.

Sample Variances

Suppose that our population consists of the numbers 1, 4, and 6.

(a) Calculate the population variance.

$$\sigma^2 = \frac{\left(1-\frac{11}{3}\right)^2+\left(4-\frac{11}{3}\right)^2+\left(6-\frac{11}{3}\right)^2}{3} = \frac{38}{9}$$

(b) Take all possible samples of size 2 (with replacement) from this population and calculate the sample variances.

Sample ($n = 2$)	1,1	1,4	1,6	4,1	4,4	4,6	6,1	6,4	6,6
s^2	0	4.5	12.5	4.5	0	2.0	12.5	2.0	0

Mean of the Sample Variances

Sample ($n = 2$)	1,1	1,4	1,6	4,1	4,4	4,6	6,1	6,4	6,6
s^2	0	4.5	12.5	4.5	0	2.0	12.5	2.0	0

(c) Calculate the mean of all the sample variances from part (b).

The table shows the probability distribution for s^2.

Notice that $\sigma^2 = \mu_{s^2}$.

s^2	$P(s^2)$	$s^2 \cdot P(s^2)$
0	$\frac{3}{9}$	0
2	$\frac{2}{9}$	$\frac{4}{9}$
4.5	$\frac{2}{9}$	$\frac{9}{9}$
12.5	$\frac{2}{9}$	$\frac{25}{9}$
Total		$\mu_{s^2} = \frac{38}{9}$

Sample Standard Deviations

Definition 7.1.9

The **sampling distribution of the sample standard deviation** is the distribution of sample standard deviations with all samples having the same size n taken from the same population.

Sample Standard Deviations

The following are true for any sampling distribution of sample standard deviations:

1. As the sample size increases, the sampling distribution of sample standard deviations becomes less skewed and more concentrated around the population standard deviation, though it is not generally normally distributed.
2. The mean of all sample standard deviations for samples of the same size taken from the same population is NOT equal to the population standard deviation. In other words, sample standard deviations do NOT *target* the population standard deviation.

Standard Deviation of Each Sample

Suppose that our population consists of the numbers 1, 4, and 6.

(a) Calculate the population standard deviation.

$$\sigma = \sqrt{\frac{38}{9}} \approx 2.0548$$

(b) Take all possible samples of size 2 (with replacement) from this population and calculate the sample standard deviations.

Sample ($n = 2$)	1,1	1,4	1,6	4,1	4,4	4,6	6,1	6,4	6,6
s	0	$\sqrt{4.5}$	$\sqrt{12.5}$	$\sqrt{4.5}$	0	$\sqrt{2}$	$\sqrt{12.5}$	$\sqrt{2}$	0

Mean of the Values of s

Sample ($n = 2$)	1,1	1,4	1,6	4,1	4,4	4,6	6,1	6,4	6,6
s	0	$\sqrt{4.5}$	$\sqrt{12.5}$	$\sqrt{4.5}$	0	$\sqrt{2}$	$\sqrt{12.5}$	$\sqrt{2}$	0

(c) Calculate the mean of all the sample standard deviations from part (b).

The table shows the probability distribution for s.

Note that, unlike the previous examples, the mean of the sample standard deviations is not the same as the population standard deviation: $1.5713 \neq 2.0548$.

s	$P(s)$	$s \cdot P(s)$
0	$\frac{3}{9}$	0
$\sqrt{2}$	$\frac{2}{9}$	≈ 0.3143
$\sqrt{4.5}$	$\frac{2}{9}$	≈ 0.4714
$\sqrt{12.5}$	$\frac{2}{9}$	≈ 0.7857
Total		$\mu_s \approx 1.5713$

Estimators

Definition 7.1.11

An **estimator** is a statistic used to infer (or estimate) the value of a population parameter.

Definition 7.1.12

An **unbiased estimator** is a statistic that targets the value of the corresponding population parameter.

Definition 7.1.13

A **biased estimator** is a statistic that does not target the value of the corresponding population parameter.

Biased or Unbiased?

In Examples 1 to 4, which statistics targeted the population parameter?

The following statistics are unbiased estimators:

- proportion
- mean
- variance

The following statistics are biased estimators:

- standard deviation
- median
- range